Raspberry Pi Camera Module Setup Guide

Introduction

This guide walks through the complete setup process for a Raspberry Pi with an IMX290/IMX462 camera module, including device tree overlay configuration, verification steps, and debugging techniques.

Prerequisites

- Raspberry Pi 4 or 5 with Raspberry Pi OS (Bookworm or later)
- IMX290 or IMX462 camera module with CSI-2 ribbon cable
- SSH access to the Raspberry Pi or direct terminal access
- Knowledge of basic command-line operations

Part 1: Initial Hardware Setup

Step 1: Connect the Camera Module

1. Power off the Raspberry Pi completely
2. Locate the Camera CSI-2 connector (between the Ethernet and USB-C ports on Pi 4, or near the IO headers on Pi 5)
3. Gently pull up the black retention clip on the CSI connector
4. Insert the ribbon cable with the blue side facing the Ethernet port (or toward the back of the board)
5. Push the retention clip down until it clicks
6. Power on the Raspberry Pi

Step 2: Enable I2C Interface

The camera communicates via I2C, which must be enabled:

sudo raspi-config

Navigate to: **Interface Options → I2C → Yes → Finish**

Or edit directly:

sudo nano /boot/firmware/config.txt

Ensure these lines are present (uncommented):

dtparam=i2c_arm=on
dtparam=i2c_vc=on

Save: `Ctrl+X` → `Y` → `Enter`

Part 2: Configure Device Tree Overlay

The device tree overlay tells the kernel how to communicate with your specific camera model.

Step 3: Edit the config.txt File

sudo nano /boot/firmware/config.txt

For IMX290 Camera

Locate the camera auto-detection line and modify:

Find:
camera_auto_detect=1

Change to:
camera_auto_detect=0

Then add to the `[all] section:`
dtoverlay=imx290,clock-frequency=74250000,cam1
gpu_mem=128

Parameters explained:

- `imx290` - Camera sensor type (use `imx462` if you have that model)
- `clock-frequency=74250000` - Clock frequency in Hz (74.25 MHz standard)
- `cam1` - Camera port (use `cam0` for single camera or alternative port)
- `gpu_mem=128` - GPU memory allocation (128 MB minimum for camera)

For IMX462 Camera

If using the IMX462 variant, the overlay is similar but the sensor name differs:

dtoverlay=imx462,clock-frequency=74250000,cam1
gpu_mem=128

Step 4: Save and Reboot

After editing config.txt:

sudo reboot

Wait for the Raspberry Pi to restart (approximately 1-2 minutes).

Part 3: Verify Camera Installation

Step 5: Check Available I2C Buses

First, verify the I2C interfaces are loaded:

ls /dev/i2c*

Expected output (may vary):
/dev/i2c-1 /dev/i2c-20 /dev/i2c-21

Different buses represent different physical I2C connections. The actual camera I2C is internal (not directly numbered in /dev).

Step 6: Scan I2C Buses (Optional - See Debugging Section)

You can attempt to scan for I2C devices:

i2cdetect -y 20
i2cdetect -y 21

Note: On modern libcamera systems (Raspberry Pi OS Bookworm+), these buses may show unusual patterns because the GPU manages camera I2C directly. See Debugging section for explanation.

Step 7: List Available Cameras (MOST RELIABLE METHOD)

Use the libcamera utility to detect connected cameras:

rpicam-hello --list-cameras

Expected output for IMX290/IMX462: Available cameras

0 : imx462 [1920x1080 12-bit RGGB] (/base/soc/i2c0mux/i2c@1/imx290@1a)
Modes: 'SRGGB10_CSI2P' : 1280x720 [60.00 fps - (320, 180)/1280x720 crop]
1920x1080 [60.00 fps - (0, 0)/1920x1080 crop]
'SRGGB12_CSI2P' : 1280x720 [60.00 fps - (320, 180)/1280x720 crop]
1920x1080 [60.00 fps - (0, 0)/1920x1080 crop]

Success indicators:

- Camera model is listed (imx290 or imx462)
- I2C address shows as `@1a` (standard I2C address)
- Resolution and frame rates are available

- Multiple color modes (SRGGB10_CSI2P, SRGGB12_CSI2P) are supported

Step 8: Test Camera Functionality

Take a test image to verify the camera works:

rpicam-still -o test.jpg

Or preview for 5 seconds:

rpicam-hello --timeout 5000

Look for a preview window (if using desktop) or confirmation that the image was captured.

Part 4: Debugging Steps

Issue: Camera Not Detected

If `rpicam-hello --list-cameras` shows no cameras:

Check 1: Verify Hardware Connection

Look for any error messages in kernel logs

dmesg | grep -i camera
dmesg | grep -i imx

If nothing appears, the kernel may not be detecting the camera hardware.

Check 2: Verify Device Tree Overlay Loaded

dtoverlay -l

Expected output should include:
imx290

If not listed, the overlay didn't load. Check config.txt for syntax errors.

Check 3: Verify GPIO and Power

Check if camera power pin is enabled

cat /proc/device-tree/base/soc/gpio@7e200000/camera_0_power/status

Should show "okay" if powered correctly.

Check 4: Verify I2C Communication

List all I2C buses and their drivers:

ls -la /dev/i2c*
i2cdetect -l

The camera should be on one of the i2c buses (typically shown as `i2c@1` internally).

Check 5: Review Full Kernel Logs

sudo journalctl -u libcamera-server -n 50

Or check for media framework messages:

dmesg | tail -20

Issue: i2cdetect Shows All Addresses Responding

This is normal on modern systems and not a failure condition.

Explanation

Buses like `/dev/i2c-20` and `/dev/i2c-21` may show all addresses responding when probed with i2cdetect. This occurs because:

1. **GPU controls camera I2C directly** - The Broadcom VideoCore GPU manages camera I2C communication, not the ARM processor
2. **Probing can interfere** - Direct I2C probing may cause the GPU to respond with all addresses
3. **Not a camera issue** - The camera is on an internal I2C bus (`i2c@1`) managed by the GPU, not on numbered buses like i2c-20

Solution: Ignore i2cdetect output for camera verification. Use `rpicam-hello --list-cameras` instead, which queries libcamera (the correct interface).

Issue: Camera Detected but Won't Capture

Check 1: Verify GPU Memory Allocation

grep gpu_mem /boot/firmware/config.txt

Should show:
gpu_mem=128

Or higher (256 for higher resolutions). Increase if needed and reboot.

Check 2: Check Camera Permissions

groups $USER

If "video" group is not listed:

sudo usermod -aG video $USER

Then log out and back in for changes to take effect.

Check 3: Test with libcamera-hello

libcamera-hello --timeout 5000

This is an alternative capture method that confirms libcamera functionality.

Issue: "Cannot open camera device"

Check 1: Verify Correct Camera Overlay

Ensure you're using the correct overlay name in config.txt:

- For IMX290: `dtoverlay=imx290`
- For IMX462: `dtoverlay=imx462`

Verify in config.txt:

grep dtoverlay /boot/firmware/config.txt

Check 2: Check for Duplicate or Conflicting Overlays

grep -i "dtoverlay.*cam" /boot/firmware/config.txt

Should only show one camera overlay. Remove duplicates or conflicting lines.

Check 3: Verify CSI-2 Connector Port

If using multiple cameras or if camera is connected to alternate CSI port, specify correctly:

dtoverlay=imx290,clock-frequency=74250000,cam0

(Change `cam1` to `cam0` or `cam1` based on physical port used)

Part 5: Advanced Configuration

Using Higher Resolution

For 4K capture (if sensor supports):

rpicam-still -o image.jpg --width 3840 --height 2160

Check supported resolutions:

rpicam-hello --list-cameras

Look for available modes in the output.

Adjusting Sensor Parameters

Exposure control

rpicam-still -o test.jpg --shutter 10000 --gain 2

White balance

rpicam-still -o test.jpg --awb daylight

Brightness/contrast

rpicam-still -o test.jpg --brightness 0.2 --contrast 1.5

Recording Video

Record 10 seconds of 1080p video at 30fps

rpicam-vid -o video.h264 --width 1920 --height 1080 --framerate 30 --timeout 10000

Troubleshooting Checklist

Use this checklist when the camera is not working:

- [ ] Hardware is securely connected (ribbon cable fully inserted)
- [ ] `sudo raspi-config` shows I2C enabled
- [ ] `/boot/firmware/config.txt` has correct `dtoverlay` line with camera model
- [ ] GPU memory is allocated (`gpu_mem=128` or higher)
- [ ] System has been rebooted since config changes
- [ ] `rpicam-hello --list-cameras` shows camera detected
- [ ] Camera address shows as `@1a`
- [ ] Sensor is listed as imx290 or imx462
- [ ] Resolutions and frame rates are available
- [ ] User is in video group (`groups $USER` includes "video")
- [ ] `rpicam-hello --timeout 5000` runs without errors

Summary of Key Commands

Enable I2C (one-time)

sudo raspi-config

Edit configuration

sudo nano /boot/firmware/config.txt

Reboot to apply changes

sudo reboot

List available cameras (primary verification)

rpicam-hello --list-cameras

Check loaded device tree overlays

dtoverlay -l

Scan I2C buses (supplementary)

i2cdetect -y 20
i2cdetect -y 21

View kernel logs for camera

dmesg | grep -i camera

Capture test image

rpicam-still -o test.jpg

Preview for 5 seconds

rpicam-hello --timeout 5000

Record video

rpicam-vid -o video.h264 --timeout 10000

References

[1] Raspberry Pi Foundation. (2024). Camera Module Configuration. https://www.raspberrypi.com/documentation/accessories/camera.html

[2] Libcamera Project. (2024). Raspberry Pi Camera Support. https://libcamera.org/

[3] Broadcom VideoCore GPU Documentation. Device Tree Bindings for IMX290 Sensor.

[4] Raspberry Pi OS Bookworm Release Notes. (2024). https://www.raspberrypi.com/news/bookworm-the-new-raspberry-pi-os/